DRONES, UAVS AND ROBOTICS POLICY

AIRCRAFT-TO-SYSTEM TERMINOLOGY LADDER → INTEGRATED UAS ARCHITECTURE → FLIGHT-CONTROL FEEDBACK LOOP → PLATFORM CATEGORY COMPARISON → HUMAN-CONTROL CONTINUUM → SWARM COORDINATION MESH

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◇ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

00

STAGE 00

AIRCRAFT-TO-SYSTEM TERMINOLOGY LADDER

AIRCRAFT-TO-SYSTEM TERMINOLOGY LADDER · AIRCRAFT ONLY · AIRCRAFT + REMOTE STATION + C2 + ASSOCIATED ELEMENTS · REMOTELY PILOTED UAS; PILOT IN COMMAND · COLLOQUIAL UMBRELLA · UNMANNED DOES NOT MEAN AUTONOMOUS

UAV → aircraft only

UAS → aircraft + remote station + C2 + associated elements → RPAS → remotely piloted UAS; pilot in command → DRONE → colloquial umbrella

STARTING CONDITION / INPUT

- UAV → aircraft only
- UAS → aircraft + remote station + C2 + associated elements

SCIENTIFIC OR ENGINEERING MECHANISM

- RPAS → remotely piloted UAS; pilot in command
- DRONE → colloquial umbrella

OUTPUT / FAILURE BOUNDARY

- RULE → unmanned does not mean autonomous
- UAV → aircraft only

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RPAS → remotely piloted UAS; pilot in command.

01

STAGE 01

INTEGRATED UAS ARCHITECTURE

INTEGRATED UAS ARCHITECTURE · AIRFRAME + POWER + PROPULSION · FLIGHT CONTROLLER + NAVIGATION SENSORS · PAYLOAD-DATA LINKS · REMOTE PILOT STATION OR ONBOARD LOGIC · MISSION CAPABILITY

AIRFRAME + POWER + PROPULSION → FLIGHT CONTROLLER + NAVIGATION SENSORS → COMMAND / TELEMETRY / PAYLOAD-DATA LINKS → REMOTE PILOT STATION OR ONBOARD LOGIC → PAYLOAD → mission capability

CONCEPT / ARCHITECTURE

- AIRFRAME + POWER + PROPULSION
- FLIGHT CONTROLLER + NAVIGATION SENSORS

MECHANISM / APPLICATION

- COMMAND / TELEMETRY / PAYLOAD-DATA LINKS
- REMOTE PILOT STATION OR ONBOARD LOGIC

READINESS / STATUS LIMIT

- PAYLOAD → mission capability
- AIRFRAME + POWER + PROPULSION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REMOTE PILOT STATION OR ONBOARD LOGIC; PAYLOAD → mission capability.

02

STAGE 02

FLIGHT-CONTROL FEEDBACK LOOP

FLIGHT-CONTROL FEEDBACK LOOP · DESIRED STATE · MEASURED STATE · COMPARE + CORRECT · CONTROL SURFACES · PHYSICAL RESPONSE · NEW STATE · SENSED AGAIN

COMMAND / PATH → desired state

IMU / SENSORS → measured state → CONTROLLER → compare + correct → ROTORS / CONTROL SURFACES → physical response

CONCEPT / ARCHITECTURE

- COMMAND / PATH → desired state
- IMU / SENSORS → measured state

MECHANISM / APPLICATION

- CONTROLLER → compare + correct
- ROTORS / CONTROL SURFACES → physical response

READINESS / STATUS LIMIT

- NEW STATE → sensed again
- COMMAND / PATH → desired state

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ROTORS / CONTROL SURFACES → physical response; NEW STATE → sensed again.

03

STAGE 03

PLATFORM CATEGORY COMPARISON

PLATFORM CATEGORY COMPARISON · HOVER + CONFINED OPERATION · ROTOR-BORNE FLIGHT · WIDER-AREA COVERAGE · LAUNCH/RECOVERY NEED · HYBRID VTOL · VERTICAL ACCESS + WING CRUISE · INTEGRATION COMPLEXITY

MULTIROTOR → hover + confined operation	rotor-borne flight
FIXED-WING → wider-area coverage	launch/recovery need
HYBRID VTOL → vertical access + wing cruise	integration complexity

SYSTEM / DEVICE / INSTITUTION

- MULTIROTOR → hover + confined operation
- rotor-borne flight
- FIXED-WING → wider-area coverage

DECISIVE TECHNICAL DISTINCTION

- launch/recovery need
- HYBRID VTOL → vertical access + wing cruise
- integration complexity

STATUS / UNIT / EVIDENCE LIMIT

- PAYLOAD CHANGES → mass + power + data + control burden
- NO LABEL → universal range/endurance guarantee

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PAYLOAD CHANGES → mass + power + data + control burden.

04

STAGE 04

HUMAN-CONTROL CONTINUUM

HUMAN-CONTROL CONTINUUM · HUMAN DIRECTLY COMMANDS · FIXED SEQUENCE · SENSES AND SELECTS ACTIONS · AI SUBSYSTEM · PERCEPTION/DECISION SUPPORT, NOT LEGAL AUTHORITY · HUMAN OVERSIGHT + DEFINED FALLBACK

CONCEPT / ARCHITECTURE

- TELEOPERATION → human directly commands
- AUTOMATION → fixed sequence

MECHANISM / APPLICATION

- AUTONOMY → senses and selects actions
- AI SUBSYSTEM → perception/decision support, not legal authority

READINESS / STATUS LIMIT

- ACCOUNTABILITY → human oversight + defined fallback
- TELEOPERATION → human directly commands

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AI SUBSYSTEM → perception/decision support, not legal authority.

05

STAGE 05

SWARM COORDINATION MESH

SWARM COORDINATION MESH · MULTIPLE AIRCRAFT · SELF AND NEIGHBOUR STATE · SHARED TASK/STATE INFORMATION · CONTROL RULES · ALLOCATION + FORMATION + ADAPTATION · FAILURE TEST · LINK LOSS OR MEMBER LOSS

MEMBERS → multiple aircraft → LOCALISATION → self and neighbour state → COMMUNICATION → shared task/state information → CONTROL RULES → allocation + formation + adaptation → FAILURE TEST → link loss or member loss

CONCEPT / ARCHITECTURE

- MEMBERS → multiple aircraft
- LOCALISATION → self and neighbour state

MECHANISM / APPLICATION

- COMMUNICATION → shared task/state information
- CONTROL RULES → allocation + formation + adaptation

READINESS / STATUS LIMIT

- FAILURE TEST → link loss or member loss
- MEMBERS → multiple aircraft

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FAILURE TEST → link loss or member loss.

06

STAGE 06

VLOS TO BVLOS ASSURANCE BRIDGE

VLOS TO BVLOS ASSURANCE BRIDGE · PILOT MAINTAINS UNAIDED VISUAL CONTACT · AIRCRAFT BEYOND VISUAL ENVELOPE · RELIABLE C2 + NAVIGATION + DETECT-AND-AVOID · LOST LINK + DIVERSION + TERMINATION LOGIC · TRIAL AUTHORISATION · ROUTINE PERMISSION

VLOS → pilot maintains unaided visual contact

BVLOS → aircraft beyond visual envelope → BRIDGE → reliable C2 + navigation + detect-and-avoid → CONTINGENCY → lost link + diversion + termination logic

CONCEPT / ARCHITECTURE

- VLOS → pilot maintains unaided visual contact
- BVLOS → aircraft beyond visual envelope

MECHANISM / APPLICATION

- BRIDGE → reliable C2 + navigation + detect-and-avoid
- CONTINGENCY → lost link + diversion + termination logic

READINESS / STATUS LIMIT

- STATUS → trial authorisation != routine permission
- VLOS → pilot maintains unaided visual contact

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONTINGENCY → lost link + diversion + termination logic.

07

STAGE 07

COUNTER-UAS DECISION FUNNEL

COUNTER-UAS DECISION FUNNEL · CLASSIFY AND MAINTAIN PICTURE · AUTHORITY + PROPORTIONALITY + SURROUNDINGS · OTHER AUTHORISED RESPONSE · JAMMING/SPOOFING CAN AFFECT LEGITIMATE SYSTEMS

CONCEPT / ARCHITECTURE

- DETECT → radar
- IDENTIFY / TRACK → classify and maintain picture

MECHANISM / APPLICATION

- DECIDE → authority + proportionality + surroundings
- MITIGATE → electronic

READINESS / STATUS LIMIT

- other authorised response
- LIMIT → jamming/spoofing can affect legitimate systems

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MITIGATE → electronic | capture | other authorised response.

08

STAGE 08

CIVILIAN APPLICATION EVIDENCE GRID

CIVILIAN APPLICATION EVIDENCE GRID · WILDLIFE OBSERVATION · DISASTER ASSESSMENT · PILOT OR AUTHORISED MISSION, NOT BLANKET SERVICE · IMAGERY + INTERPRETATION + FIELD VERIFICATION

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
AGRICULTURE → monitor / map / spray	GOVERNANCE → survey / inspection / disaster assessment	EVIDENCE → imagery + interpretation + field verification
SCIENCE → volcano / wildlife observation	LOGISTICS → pilot or authorised mission, not blanket service	AGRICULTURE → monitor / map / spray

CONCEPT / ARCHITECTURE

- AGRICULTURE → monitor / map / spray
- SCIENCE → volcano / wildlife observation

MECHANISM / APPLICATION

- GOVERNANCE → survey / inspection / disaster assessment
- LOGISTICS → pilot or authorised mission, not blanket service

READINESS / STATUS LIMIT

- EVIDENCE → imagery + interpretation + field verification
- AGRICULTURE → monitor / map / spray

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE → imagery + interpretation + field verification.

09

STAGE 09

SAFETY PRIVACY AND LIABILITY SHIELD

SAFETY PRIVACY AND LIABILITY SHIELD · AIRWORTHINESS + MAINTENANCE + FAILSAFES · PILOT + WEATHER + ROUTE + PAYLOAD DISCIPLINE · PURPOSE + ACCESS + RETENTION + GEOSPATIAL SENSITIVITY · C2/NAVIGATION/PAYLOAD-LINK RESILIENCE · TRACE OPERATOR/SOFTWARE/MANUFACTURER FUNCTION · MUST REMEMBER · DRONES COMBINE AIRFRAME, PROPULSION, CONTROL, COMMUNICATION/NAVIGATION

ENGINEERING → airworthiness + maintenance + failsafes → OPERATIONS → pilot + weather + route + payload discipline → DATA → purpose + access + retention + geospatial sensitivity → CYBER → C2/navigation/payload-link resilience → LIABILITY → trace operator/software/manufacturer function

CONCEPT / ARCHITECTURE

- ENGINEERING → airworthiness + maintenance + failsafes
- OPERATIONS → pilot + weather + route + payload discipline

MECHANISM / APPLICATION

- DATA → purpose + access + retention + geospatial sensitivity
- CYBER → C2/navigation/payload-link resilience

READINESS / STATUS LIMIT

- LIABILITY → trace operator/software/manufacturer function
- MUST REMEMBER: Drones combine airframe, propulsion, control, communication/navigation,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Drones combine airframe, propulsion, control, communication/navigation,.

10

STAGE 10

INDIAN GOVERNANCE RESPONSIBILITY MAP

INDIAN GOVERNANCE RESPONSIBILITY MAP · CIVIL REGULATION · AIRSPACE AND TRAFFIC SERVICES · AVIATION SECURITY · DIGITAL SKY · WORKFLOW PLATFORM, NOT ATC · CLOSE DISTINCTION · DRONE IS NOT NECESSARILY AUTONOMOUS, REMOTE PILOTING IS NOT

MOCA → policy / notification

DGCA → civil regulation / certification → AAI / ATC → airspace and traffic services → BCAS → aviation security

CONCEPT / ARCHITECTURE

- MOCA → policy / notification
- DGCA → civil regulation / certification

MECHANISM / APPLICATION

- AAI / ATC → airspace and traffic services
- BCAS → aviation security

READINESS / STATUS LIMIT

- DIGITAL SKY → workflow platform, not ATC
- CLOSE DISTINCTION: Drone is not necessarily autonomous, remote piloting is not AI...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Drone is not necessarily autonomous, remote piloting is not AI..

11

STAGE 11

APPROVAL-TO-OUTCOME STATUS RAIL

APPROVAL-TO-OUTCOME STATUS RAIL · LEGAL TEXT · TYPE CERTIFICATE · DISTINCT COMPLIANCE CONDITIONS · AUTHORISED OR SUPPORTED ROUTE · NOT OPERATIONAL DEPLOYMENT · NOT VERIFIED SAFETY, SCALE OR OUTCOME · EVIDENCE LIMIT

RULE / AMENDMENT → legal text → TYPE CERTIFICATE / UIN / RPC → distinct compliance conditions → TRIAL / SCHEME / INCENTIVE → authorised or supported route → PROCUREMENT / DELIVERY → not operational deployment → DEPLOYMENT → not verified safety, scale or outcome

CONCEPT / ARCHITECTURE

- RULE / AMENDMENT → legal text
- TYPE CERTIFICATE / UIN / RPC → distinct compliance conditions

MECHANISM / APPLICATION

- TRIAL / SCHEME / INCENTIVE → authorised or supported route
- PROCUREMENT / DELIVERY → not operational deployment

READINESS / STATUS LIMIT

- DEPLOYMENT → not verified safety, scale or outcome
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State platform class, weight/category,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State platform class, weight/category,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State platform class, weight/category,.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · THE MOST IMPORTANT ANALYTICAL SPLIT IS BETWEEN AIRFRAME MANUFACTURING · THE LATTER DEPENDS ON SENSORS, SOFTWARE, COMMUNICATIONS AND PAYLOAD · DRONE POLICY SITS AT THE INTERSECTION OF AVIATION LAW · ROBOTICS CAPABILITY IS INCREASINGLY DUAL-USE · AGRICULTURE SPRAYING AND LOGISTICS CAN COEXIST WITH DEFENCE-SUPPORT OR · RULE-MAKING AND SCHEME-LEVEL ANCHOR. · COMPLIANCE AND CIVIL-DRONE REGULATORY AUTHORITY

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: The most important analytical split is between airframe manufacturing and mission capability
- the latter depends on sensors, software, communications and payload integration.
- CAUTION: Drone policy sits at the intersection of aviation law, digital governance, privacy, internal security and industrial policy.
- CAUTION: Robotics capability is increasingly dual-use: agriculture spraying and logistics can coexist with defence-support or surveillance relevance.

READINESS, UNIT OR STATUS LIMIT

- MoCA: rule-making and scheme-level anchor.
- DGCA: compliance and civil-drone regulatory authority.
- Digital Sky: public-facing airspace and permission architecture.
- PLI for drones and drone components: industrial push for indigenous ecosystem capability.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- Training/certification ecosystem: necessary for safe scaling of operations.
- Agriculture: precision application and monitoring can reduce time and improve coverage.
- Disaster response: rapid imaging and access to difficult terrain improves first-response intelligence.
- Surveying/logistics/infrastructure inspection: drones reduce time and increase reach.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels